

Madan Bhandari College of Engineering
Urlabari-3, Morang

Final Assessment- 2025, Fall

Level: Bachelor

Programme: BE(Computer)

Course: Dat Science and Analytics

Year/Part:IV/I

Full Marks : 100

Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

- ✓ 1. (a) Explain the Knowledge Discovery in Database (KDD) process with a neat diagram. Discuss each stage in detail. [7]

- ✓ (b) A numeric variable has the following values:
12.5, 45.6, 78.9, 23.4, 56.7, 89.1, 34.2, 67.8

- i. Standardize the data.
ii. Construct a histogram and comment on distribution shape. [8]

OR

Why Use a Data Analysis Pipeline? Differentiate between Structured and unstructured data in appropriate examples.

- ✓ 2. (a) Explain power-law distribution with real-life examples. Why is it called heavy-tailed distribution? [8]

OR

Explain empirical distribution for numeric and categorical data. Discuss normal, exponential and binomial distributions.

- ✓ (b) The following data represents age (X) and annual income (Y in thousands):

X	22	25	30	35	40
Y	25	30	45	50	60

- i. Compute Pearson correlation coefficient.
ii. Interpret the result.
iii. Test significance using t-test. [7]

3. (a) What is Ordinary Least Squares (OLS)? Derive the OLS estimator using matrix formulation. [8]

(b) You are given:

Obs	X	Y
1	1	3
2	2	5
3	3	7
4	4	9
5	5	11

- i. Estimate regression coefficients.
ii. Compute R^2 and Adjusted R^2 .
iii. Predict Y when $X=6$. [7]

4. (a) Derive the Nadaraya-Watson kernel regression estimator. Discuss bandwidth selection methods. [8]

OR

Explain K-means clustering algorithm. Discuss distance measures for mixed-type data; [8].

- (b) Given dataset with 3 variables and 5 observations, explain how PCA is computed using eigenvalue decomposition. How do you determine variance explained? [8]

5. (a) At a node of a decision tree, the following data is observed:

A	B	Class
T	T	Yes
T	F	Yes
F	T	No
F	F	No
T	T	Yes
F	F	No

Using Gini index, determine the best attribute for splitting. [8]

- (b) Explain CART algorithm for classification trees. [7]

6. (a) Explain ARMA and ARIMA models with mathematical formulation. [7]

(b) How do you select optimal lag length using AIC and BIC? Explain with formula. [7]

7. Write short notes on (Any Two): [2×5=10]

- a) Multicollinearity and VIF b) Logistic Regression
c) Causation vs Correlation

*****All The Best*****

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

1. (a) Explain data preprocessing techniques:

- i. Data cleaning
- ii. Data integration
- iii. Data transformation
- iv. Discretization [8]

(b) A numeric variable has the following values:

12.4, 25.6, 31.2, 45.8, 28.9, 39.5, 41.3, 22.7, 18.6, 35.4

i. Standardize the data.

ii. Construct a histogram and comment on distribution shape. [8]

OR

Why Use a Data Analysis Pipeline? Differentiate between Structured and unstructured data in appropriate examples.

2. (a) Define numeric and categorical variables and give 3 examples of each from real-life datasets. Describe the differences between normal, exponential, and power law distributions with examples. [8]

OR

Cross-tabulation example: 20 people surveyed for Gender (M/F) and Preference (Tea/Coffee): [8]

	Tea	Coffee
Male	5	5
Female	3	7

a) Create the cross-tabulation table.

b) Compute the chi-squared value to test independence.

(b) The following data represents age (X) and annual income (Y in thousands):

X	22	25	30	35	40
Y	25	30	45	50	60

i. Compute Pearson correlation coefficient.

ii. Interpret the result. iii. Test significance using t-test. [7]

3. a) Explain R^2 and Adjusted R^2 in model evaluation. Discuss multicollinearity and Variance Inflation Factor (VIF). [8]

b) You are given the following data:

x	0.5	1.0	1.5	2.0
y	1.2	2.3	3.1	3.8

Estimate the value of y at $x = 1.2$ using the Nadaraya-Watson estimator. Take bandwidth $h = 0.3$ and Gaussian kernel. [7]

OR

Given a dataset with n variables, explain how Principal Component Analysis (PCA) is computed. How is the variance explained by each component determined? [7]

4.a) Consider a simple linear regression model: [8]

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Given:

X	1	2	3	4	5
Y	3	5	7	9	11

i) Write the matrix form of the regression model.

ii) Estimate β_0 and β_1 . iii) Predict Y when $X = 6$.

b) Describe the CART (Classification and Regression Trees) algorithm in detail. Explain how the Gini Index is used as a splitting criterion and how CART handles categorical variables. [7]

5. (a) Explain permutation-based variable importance and partial dependence plots. [8]

(b) Perform one step of K-means clustering for the following data with $k=2$. Points: (1,2), (2,1), (8,9), (9,8) Initial centroids: (1,2) and (8,9). [7]

6. (a) What is time series analysis? Explain the concepts of autocorrelation and stationarity, and discuss why stationarity is important in time series modeling. [7]

(b) What is Granger causality? What is a Causal Directed Acyclic Graph (DAG)? Explain its main components and how DAGs help in representing causal relationships. [8]

OR

Explain the importance of selecting an optimal lag length in time series models. Discuss how Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used for this purpose. [8]

7. Write short notes on (Any Two): [2×5=10]

a) Statistical significance (p-value, t-value) b) Logistic Regression

c) Causation vs Correlation

*****All The Best*****